

Soorya Narayanan V

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Graduate ML Researcher at USC working on Google and Verizon funded projects, with experience in Building Data Pipelines, Edge Deployment, Training Models, and maintaining 3 active AI systems with live user bases.

EDUCATION

University Of Southern California

Masters in Applied Data Science. GPA: 3.85/4.0

Los Angeles, CA

Expected May 2027

- **Coursework:** Data Mining (PhD), Machine Learning for Data Science, Applied Natural Language Processing.
- **Grader** for USC's Data Visualization Course (DSCI 554) – [React, Vue.js, D3.js]

Anna University

Bachelors in Artificial Intelligence and Data Science. GPA: 3.7/4.0

Chennai, India

May 2025

- **Coursework:** Data Structures & Algorithms, Machine Learning, Deep Learning, Data Science.
- **Minors:** Business Data Analytics

Awards & Certifications: <https://inkedsin.wordpress.com/soorya-narayanan-v>

WORK EXPERIENCE

Directed ML Researcher – WiDES lab (Verizon)

University of Southern California,

February 2026 – Current

- Trained PMNet CNN encoder-decoder model for 5G pathloss prediction across 5 urban scenes on 19,000+ geospatial samples; scaled training to multi-node DDP (SLURM + torchrun, FP16 mixed precision) on HPC, achieving 97% pathloss prediction accuracy and zero-shot cross-scene generalization to held-out environments.
- Engineered end-to-end ETL pipeline fusing ray-tracing simulation outputs (Wireless InSite 4.0 + NVIDIA Sionna RT) with 3D urban geometry into ML-ready pathloss map datasets; reduced test-set MAE by 57% over non-deep-learning baseline and delivered transmitter placement recommendations directly to Verizon.

Graduate Research Assistant – CeNEC lab (Google)

University of Southern California,

September 2025 – April 2026

- Architected a systematic LLM evaluation framework benchmarking “metaphor comprehension” and “auditory imagery perception” baselines across ~9 major frontier models, uncovering a ~14% confidence-accuracy gap. Building ground-truth benchmark.

ML Engineer (Model Evaluation)

Mirador Technologies,

July 2024 – August 2025

- Rebuilt Wild Face Recognition system on real-time live video streams, boosting accuracy from 42% to 93% to recognize faces accurately under ~70ms in real time, by re-architecting the data pipeline on embedding, preprocessing, and async messaging layer (RabbitMQ), use of FAISS vector database.
- Benchmarked 14+ open-source face detection and recognition models with systematic **latency-accuracy-cost** tradeoff analysis, fine-tuned selected models to meet client requirements and led technical demos, converting 2 prospects into clients.

Data Science Intern

Vizza Insurance Broking Services Pvt Ltd,

May 2023 – August 2023

- Engineered production-grade LightGBM lead-scoring model surfacing 7 KPIs from raw CRM data, cutting lead-to-conversion time from ~3 days to ~7 hours (71% reduction), and improving sales conversion by 65%, with automated retraining triggered deployment on incoming data for the sales operation team.

PROJECTS

Intelligent Bistro

May 2025

Built AI-powered food ordering app (Expo React Native + Node.js API) with 9 custom tool-calling functions, Zod validation, MCP server for production analysis, CI/CD Testing 236 tests – 94% coverage, voice TTS/STT, 4-signal recommendation engine (popularity, session, pairings, compatibility), SQLite backed order management. ([Demo Link](#))

Immunity – (LLM Routing + Edge Deployment)

January 2026 – Current

Built Clinical AI allergy tracker with LLM routing layer delegating between AWS Nova Micro and MedGemma 4B, 4-bit quantization, storing conversation in DynamoDB, and edge deployment with ONNX for active users in production.

RL from Verifiable Rewards (RLVR) – (LoRA + 4bit Quantization)

March 2025

Applied GRPO via HuggingFace TRL to fine-tune Qwen2.5-7B-Instruct on the tau2-bench; emitting structured JSON tool-call sequences using dual reward functions (format + correctness), improving JSON formatting accuracy to 87% with 4-bit NF4 quantization + LoRA (rank 16). ([GitHub](#))

Sentiment-Driven Portfolio Optimizer

March 2025 – July 2025

Combined Black-Litterman optimization with FinBERT derived sentiment signals from financial news, achieved a Sharpe Ratio of 1.43, portfolio alpha of 6.2%, and VaR (95%) of 3.1%, outperforming passive S&P 500 allocation.

SKILLS

Languages & Fundamentals: Python, Typescript, SQL, C++, Git, Linux Bash, Scala

ML Frameworks: PyTorch, TensorFlow, Keras, OpenCV, Scikit-Learn, JAX, NumPy, Huggingface, CUDA, RAG, MCP, DDS

MLOps: Flask, Docker, Kubernetes, Spark, Airflow, GitHub Actions, CI/CD Testing, Automated Retraining, RabbitMQ, Kafka.

Machine Learning: LMM, VLM, LLM, Train, Post-Training, Inference, SVD, XGBoost, NLP, CV, SFT, LoRA, Distillation.